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⚛️PHYS001: Basics of atomic physics and fission.

In this short thread-course, I'm going to get you up to speed with the basics of atomic physics.
If you already understand the words "chain reaction", "mass defect" and "fission", you can skip this course.
Everyone else, let's go!

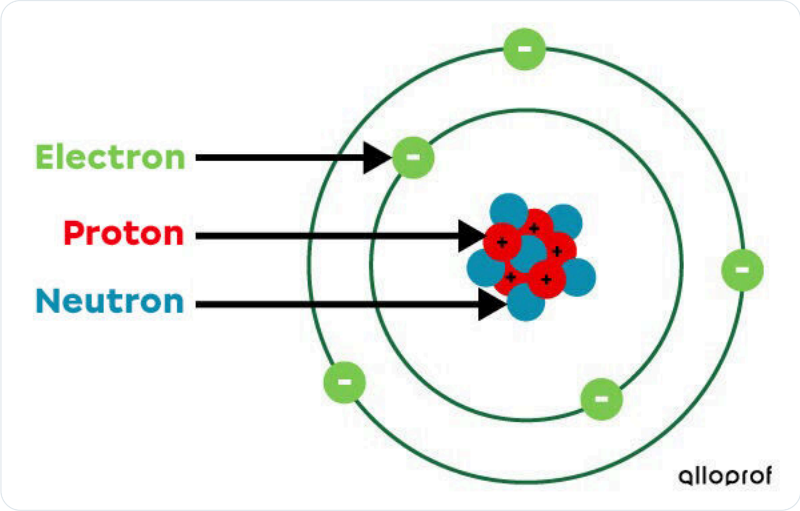
As you all know, we're made of bundles of atoms and energy!

If you ignore a lot of schizo standard model crap, there's just four basic ingredients (and only 3 in some models but we'll get to that in the future):

Electrons (e-)
Photons (γ)

Nucleons:
Protons (p+)
Neutrons (n)

Electrons and Protons are charged particles, that attract each other, and are repelled by themselves.
An atom is made up of a positively charge nucleus and a cloud of negatively charged electrons.
Photons mediate energy transitions for both the nucleus and the electron cloud.



Inside the electron cloud, chemistry happens!
Conventional explosives activate through transformation from one molecule (collection of atoms) to another, generating heat (photons) and momentum in the process.
That energy activates more reactions.

This is called a chain reaction.

An Analysis of the Explosion Limits of Hydrogen/Oxygen Mixtures with Nonlinear Chain Reactions

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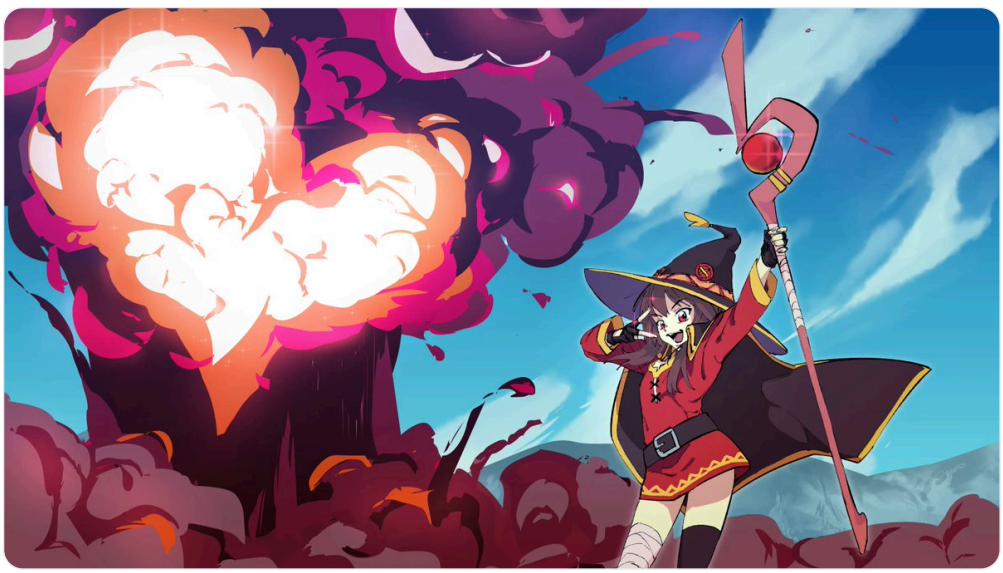
Abstract

The ignition boundary of hydrogen/oxygen mixtures is a Z-shaped curve in the pressure-temperature space, demonstrating the existence of three explosion limits. In this study, a general analysis governing all the three explosion limits in an isothermal environment is performed by considering both linear chain reactions (reactant-radical reactions) and nonlinear chain reactions (radical-radical reactions), in addition to the zeroth-order reactant-reactant reactions. For the nonlinear reactions, it is further shown that the linear-nonlinear coupling has the major influence, while the ef-

To state another way, (exothermic) reactions have two sides:

Reactant + Activator -> Product(s) + Energy

In the special case when the product of a reaction is also its activator, you get a chain reaction, since the reaction can continue until you're out of available reactants.



These conventional explosions will move the nucleus around but won't make it undergo a reaction of its own. There's a good reason for this -- it's the huge energy required! [*except LENRs]

Let's look first at the scale of energies involved in chemical reactions.

The energy it takes to strip an electron from a Hydrogen atom (which is simply a proton nucleus that has an electron spinning around it) is around 13.6 eV.

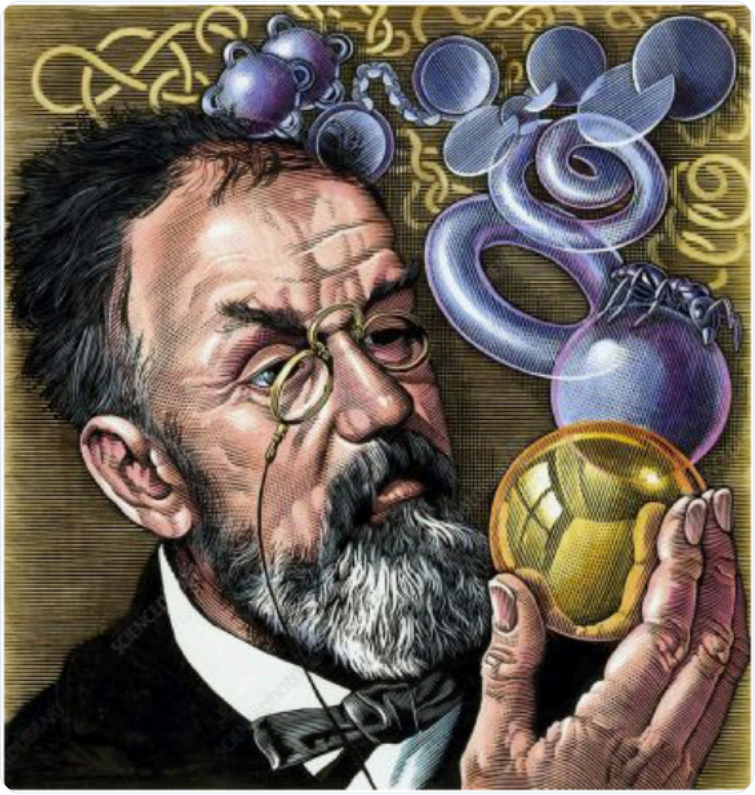
That's around 1.3 megajoules per mol.

This is a huge amount of energy, given a mol of hydrogen is about a gram...

But nuclear reactions are on a totally different level. To understand why, we have to enter mass-energy equivalence.
Poincare (not Einstein), proposed the following mass-energy relationship:

E=m0 c^2

He derived this using Lorentz transformations and self-interacting fields.

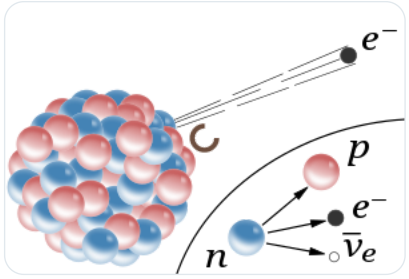


What does this mean in practice? It means that the mass of the nucleus (and indeed electrons) can get involved in reactions and produce energy.

In short: By measuring the weight of the products and reactants, we can calculate the resultant energy. Let's look at one example.

It is said that the currently modelled neutron decays into an electron and proton when isolated from the nucleus of an atom. How much energy is released?

For reference, the energy equivalence of our particles follow:
p+ = 938.28 MeV (million eV)
n = 939.57 MeV
e- = 0.511 MeV



Thus:
Mass defect = Reactant - Product = n - (p+ + e-)
= 939.6 - (938.3 + 0.511) = ~0.78 MeV

That's more than 58,000 times the energy it takes to strip an electron from a hydrogen atom!!!

We see this nuclear reaction is 4-5 orders of magnitude larger than chemical ones.

Note for the nitpickers: The electron stripping from a hydrogen example is in fact much larger than the energy released by chemical reactants that can undergo a chain reaction (explosives), so it is a sufficient example for our intentions.

Another slight detour: As it happens, these large numbers are really inconvenient for lazy bakas like me (aka physicists), and we like to use "atomic units" (based on Carbon-12, what you're made of).

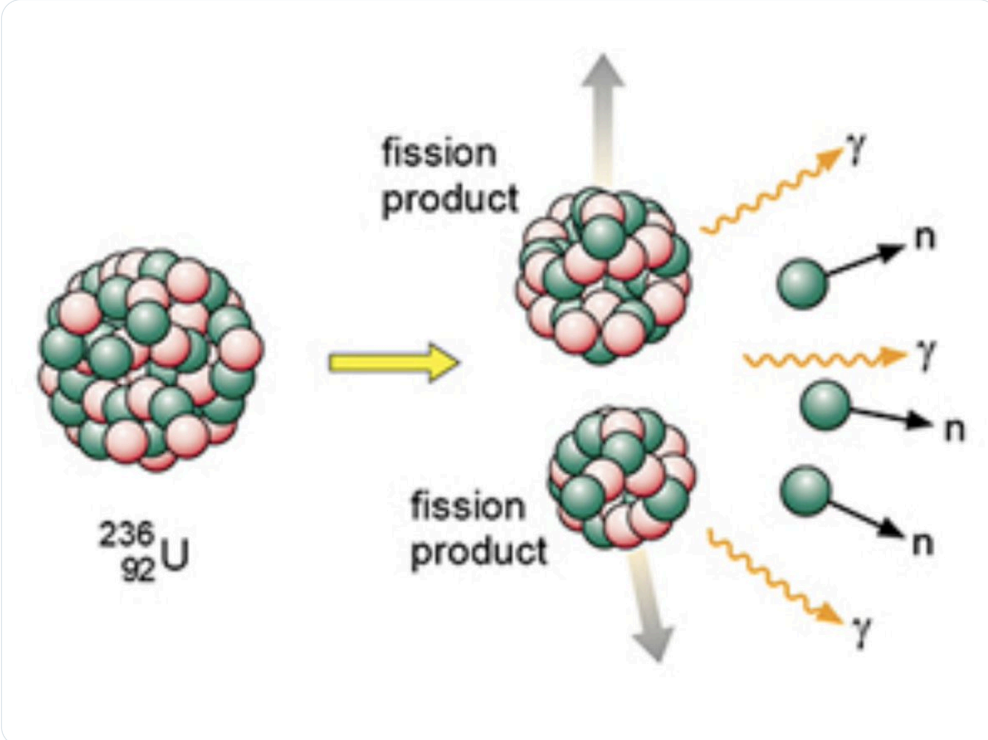
p+ = 1.007276 u
n = 1.008664 u
e- = 0.00054858 u (smol)

Now, let's move on to something more... explosive! Fission.

What is fission? Fission is when an unstable* atom splits into smaller atoms and releases various other stuff, mostly neutrons and lots of energy (photons).

* Why atoms are unstable is sort of an open question.

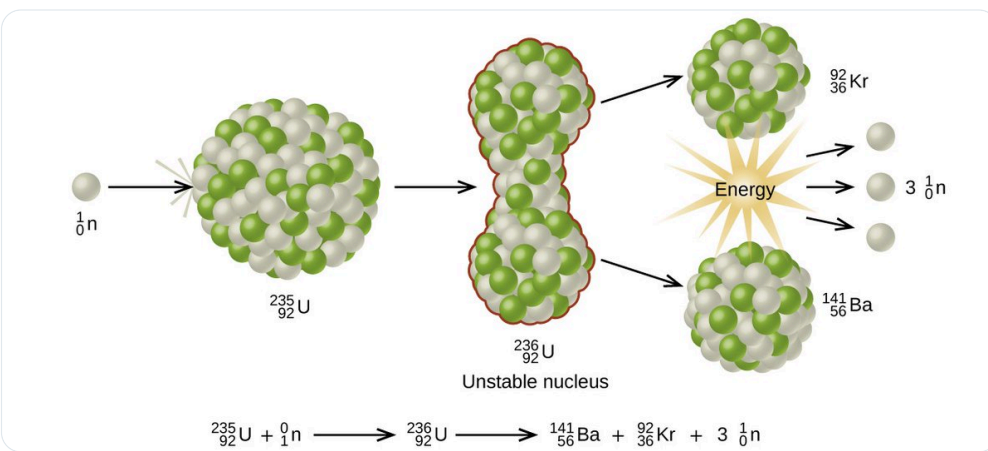
Some atoms configurations (isotopes) just don't like existing!
Take Uranium-236 for example. This is one annoyingly grumpy molecule! Give it a long enough time and it'll split into two atoms (called daughters,get it lol), while producing a lot of energy and two or three neutrons.



The energy released is immense but depends on which products are made as per the mass-defect.
How do we exploit this, if we want that juicy energy produced out of it?
Let's knock on the door of its thinner brother Uranium-235. This one has one less neutron than U-236...

What if a slow neutron were to visit U-235 and get swallowed up by it? It'll turn into a very grumpy ("excited"/deformed) U-236*. This will undergo fission almost immediately!

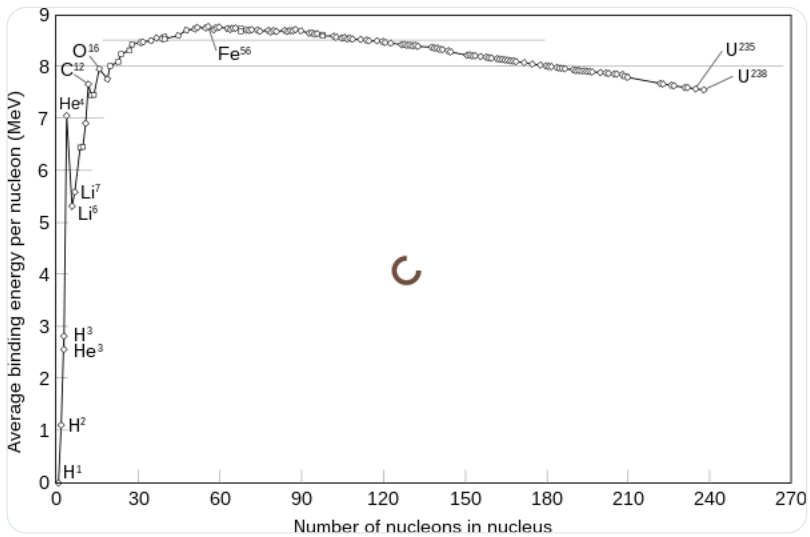
Notice that the products include an activator (a neutron). A massive chain reaction is possible!



Let's do some math on one example of our grumpster creating two Barium-141 and Krypton-92 daughters:

U-236* -> Ba-141 + Kr-92 + 3n

The atomic weights balance out but the energy produced by fission is MANY times even greater than the energy produced by spontaneous neutron decay.
[Aside: We can't have a proton visit U-235, since it will be repelled by the positively charged nucleus!]
[Also: More on why it needs to be slow later]
[Also: The origin of the binding energy associated with an atom vs its individual particles is an open problem]

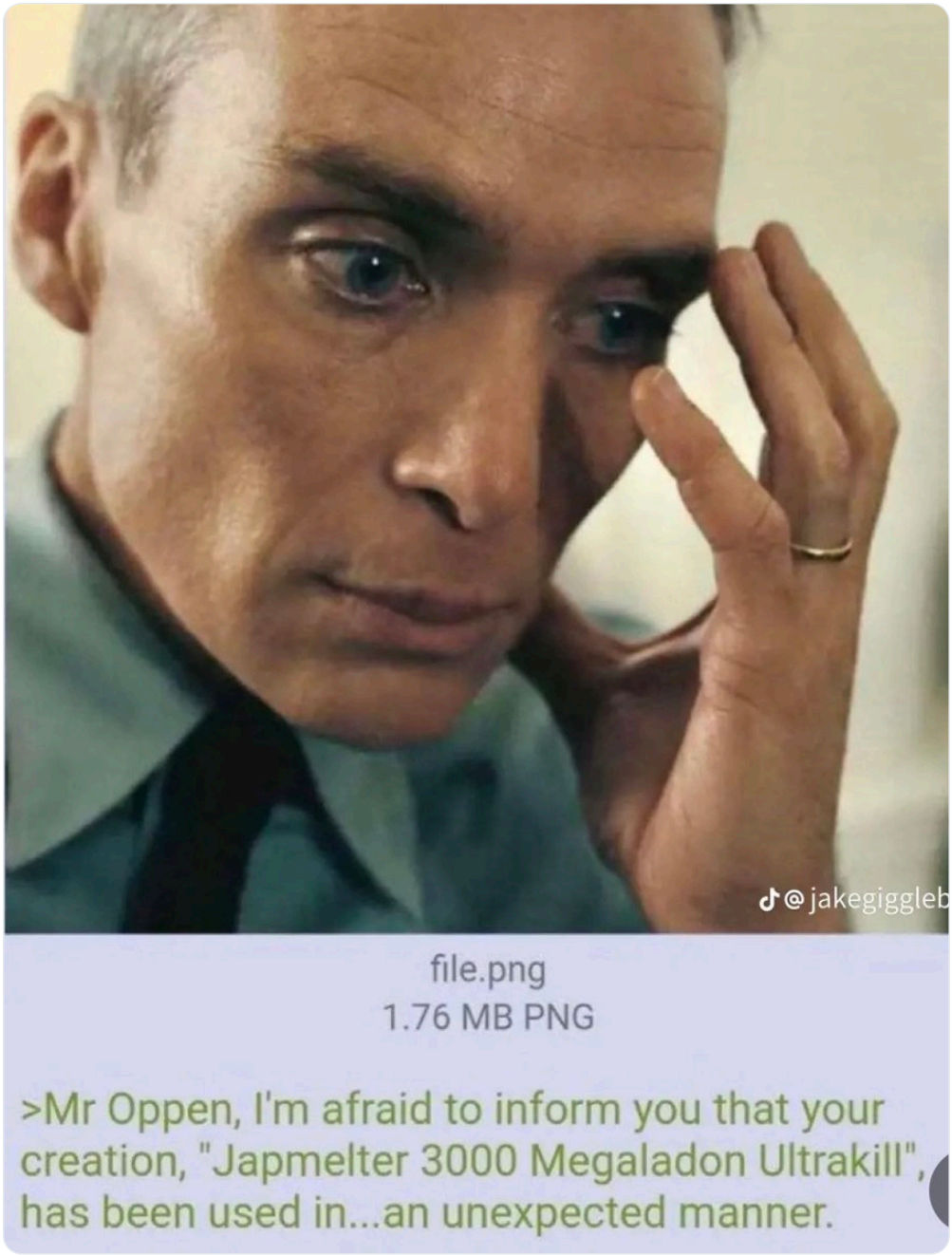


U-236* = 236.04556
Ba-141 = 140.91441
Kr-92 = 91.92615
3*n = 1.00866

Mass defect=236.045568 - (140.914411 + 91.926156 + 3*1.008664)=0.179009 u
This is equal to 166 MeV!! Great than 200 times more energetic than spontaneous neutron decay.

This mean that such a reaction, per unit, is more than 10 million times more powerful than a chemical reaction.
This is exactly why the Manhattan project was formed.

You see, conceptualising this reaction and doing the math is easy.
But turning it into a weapon is an art form.



How do we go about turning the fission idea into a weapon then?
Do we do away with our chemistry set all together? Or will chemical explosives come handy?
Where are we going to find these neutrons that keep decaying on us?

Find out next time in PHYS002! Thank you for reading 💙

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